90% of AI Engineer Interviews in 2026 = These 12 Concepts

If you're preparing for AI Engineer roles in 2026, here's the reality:

Most interviews don't test random theory anymore.
They repeatedly circle around the same **12 core concepts** — just applied in different ways.

Master these, and you're already ahead of 90% of candidates.

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1. Prompt Engineering

This is the **first layer of control** over any LLM.

You're not just asking questions — you're designing instructions.

- Techniques:
  - Zero-shot
  - Few-shot
  - Chain-of-Thought (CoT)
  - Tree-of-Thought (ToT)
  - ReAct

👉 Think of it like giving a chef a **perfect recipe with plating instructions**.

Pros:

- Instant results
- No infrastructure required

Cons:

- Fragile (breaks with small changes)
- Depends heavily on model behavior

Best Use Cases:

- Prototypes
- Chatbots
- Internal tools

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2. RAG (Retrieval-Augmented Generation)

LLMs don't "know everything." RAG fixes that.

Instead of relying on memory:

- Retrieve relevant data
- Inject it into the prompt
- Generate grounded answers

👉 Like a chef checking trusted recipe books before cooking.

Pros:

- Up-to-date knowledge
- Reduced hallucinations

Cons:

- Retrieval quality = everything

Best Use Cases:

- Enterprise Q&A
- Document search
- Support bots

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3. Vector Embeddings & Vector Databases

This is what powers **semantic search**.

Tools like:

- Pinecone

- Weaviate
- PGVector

convert text into numerical representations.

👉 Like turning every recipe into a **secret code** to find similar dishes instantly.

Pros:

- True semantic understanding

Cons:

- Expensive indexing
- Needs freshness management

Best Use Cases:

- Search
- RAG pipelines
- Recommendations

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4. Agentic AI & Tool Calling

AI is no longer just answering — it's **acting**.

Agents can:

- Plan tasks
- Use APIs/tools
- Reflect and improve

👉 Like a smart sous-chef who:

- Orders ingredients
- Adjusts cooking
- Fixes mistakes

Pros:

- Handles complex workflows

Cons:

- Can hallucinate actions
- Risk of infinite loops

Best Use Cases:

- Automation systems
- Research agents
- Multi-step workflows

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5. Chain-of-Thought & Advanced Reasoning

Instead of jumping to answers, models:

- Think step-by-step
- Reflect
- Self-correct

👉 Like a chef explaining every decision before plating.

Pros:

- Better logical accuracy

Cons:

- Higher token cost
- Slower responses

Best Use Cases:

- Math problems
- Planning
- Complex reasoning

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6. Memory Persistence & Context Management

AI systems need memory to feel “intelligent.”

Types:

- Short-term (context window)
- Long-term (vector DB, summaries)

👉 Like a smart pantry that:

- Tracks ingredients
- Suggests substitutions

Pros:

- Stateful conversations

Cons:

- Token limits
- Memory drift

Best Use Cases:

- Assistants
- Long sessions
- Agents

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7. Streaming & Async Patterns

Modern AI apps don’t wait — they **stream**.

- Token streaming
- Async tool execution
- Background processing

👉 Like serving dishes as soon as they’re ready.

Pros:

- Great user experience
- Higher throughput

Cons:

- Complex engineering

Best Use Cases:

- Chat apps
- Real-time systems

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8. Inference Optimization

This is where real engineering begins.

Techniques:

- Quantization
- Distillation
- Batching
- Caching
- vLLM

👉 Like optimizing a kitchen to serve 1000 customers fast.

Pros:

- 5–10x cheaper
- Faster responses

Cons:

- Slight quality loss

Best Use Cases:

- Production systems

- High-scale apps

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9. Token & Cost Optimization

AI isn't expensive — **bad architecture is**.

Focus on:

- Prompt compression
- Caching
- Model routing
- FinOps

👉 Like tracking every gram of ingredient in a kitchen.

Pros:

- Massive cost savings

Cons:

- Requires constant monitoring

Best Use Cases:

- High-volume applications

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10. Fine-Tuning (LoRA / QLoRA / PEFT)

When prompting isn't enough → you customize the model.

👉 Like training a chef in your restaurant's secret recipes.

Pros:

- Domain-specific performance
- Better tone & style

Cons:

- Requires data + GPUs
- Needs evaluation

Best Use Cases:

- Vertical AI apps
- Branding & tone control

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11. LLM Evaluation & Metrics

If you can't measure it, you can't improve it.

Common approaches:

- RAGAS
- LLM-as-a-judge
- Hallucination scoring
- Golden datasets

👉 Like a strict tasting panel with scorecards.

Pros:

- Objective improvements

Cons:

- Not fully automated

Best Use Cases:

- Iteration
- Monitoring
- A/B testing

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12. MLOps & Production Deployment

This is where most candidates fail.

Real-world AI =

- Serving
- Monitoring
- Drift detection
- Rollbacks
- Guardrails

👉 Like running a 24/7 professional kitchen.

Pros:

- Reliable systems

Cons:

- Heavy infrastructure work

Best Use Cases:

- Customer-facing AI
- Enterprise systems